

WU JINJUN

The Hong Kong University of Science and Technology, Hong Kong
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EDUCATION

B.Eng in Computer Science

September 2025 -

The Hong Kong University of Science and Technology

CGPA: **3.99/4.3** (top 2% in HKUST)

Relevant Coursework: Object-Oriented Programming and Data Structures, Intro to Python, C++, Intro to Artificial Intelligence, Discrete Mathematics, Computer Organization, Operating Systems

HONORS & AWARDS

Dean's List, School of Engineering

2025-26 Fall, Spring

University's Scholarship Scheme for Continuing Undergraduate Student

2026

RESERACH INTERESTS & EXPERIENCE

Interests	Reinforcement Learning (RL) Machine Learning (ML) Graphs
Experience	NaN

PROJECTS

EdgeHealth-ML: On-Device Multimodal Activity Recognition

June 2026

Tools: *Python, PyTorch, torchvision, Claude Code.*

- Designed a **dual-branch multimodal fusion architecture**: Vision branch uses MobileNetV3-Small for STFT spectrogram features; IMU branch uses a 3-layer 1D-CNN to process raw 6-channel accelerometer + gyroscope signals. Features are fused via concatenation + 2-layer MLP with dropout.
- Evaluated on the UCI HAR Dataset (30 subjects, 6 activities: walking, walking upstairs/downstairs, sitting, standing, laying), achieving 87.6% test accuracy of dynamic activities (walking 100%, stair-climbing 96%).

Kaggle Competition: Stellar Class Classification

May 2026

Tools: *Python, NumPy, Pandas, Scikit-Learn, Matplotlib.*

- Participated in a classification competition to predict galaxy and star types from 12-dimensional photometric data (redshift, spectral features, galaxy population, etc.). Performed data preprocessing including **categorical variable encoding and min-max normalization** to $[0, 1]$.
- Implemented multiple models: **Decision Stump** (from scratch), **KNN and K-means** (via Scikit-Learn). Evaluated model performance through confusion matrix, Precision, Recall, and F1 scores; identified limitations in separating overlapping galaxy and star clusters. Achieved 94% classification accuracy on the test set using KNN.

COMP3511 PA: Myshell - Simplified Linux Shell

September 2026

Tools: *C, POSIX, Multiprocess Programming, Signal Handling, File Descriptor Redirection*

- Implemented a classic fork-exec command engine: parent parses input, spawns a child via `fork()` , child loads external programs with `execvp()` , parent reaps all children via `wait()` — zero zombie processes.
- Built I/O redirector with one-pass parameter scanning for `<` and `>` tokens; used `open()` with `O_RDONLY` / `O_WRONLY` / `O_CREAT` / `O_TRUNC` flags followed by `dup2()` onto `STDIN_FILENO` / `STDOUT_FILENO` ; supports arbitrary redirect ordering. Designed a multi-stage pipeline scheduler : pre-allocated `N-1 pipe()` s, forked one child per segment stitched via `dup2()` , and explicitly closed all inherited pipe FDs across every child and the manager to eliminate classic EOF-blocked pipeline deadlocks — reliably handles up to 8-stage pipelines with no hangs or leaks.

- Implemented `exit` / `cd` as parent-executed builtins (before `fork()` , since subshell `chdir()` cannot mutate the parent's `cwd`). Registered a custom `SIGINT` handler for `Ctrl-C`. Code compiles cleanly under `gcc -std=c99 -pedantic -Wall -Wextra`, passing all 10 public + 6 hidden autograder cases.

PRACTICAL SKILLS

Programming: Python, C++

Machine Learning & Deep Learning Framework: Scikit-Learn, TensorFlow, PyTorch

Data Preprocessing & Visualization: NumPy, Pandas, Matplotlib

Operating System: Linux

Type Setting: L^AT_EX

Languages: Mandarin (native), English (IELTS 7.0)

TEACHING SERVICES

COMP2011: Programming with C++

Spring 2027 (Pending)